

9. Data logging, file layout, and capture budget

Save every frame raw (.mc), with one manifest row per frame, so that any analysis can be re-run without re-capturing. The manifest records the read-back robot pose, not only the commanded one.

Folder layout (under Noise Estimation Data/):

Characterization_<YYYYMMDD>/ sensor_config.json registration.json targets_asbuilt.csv
manifest.csv session_log.md environment_log.csv 00_registration/ A_noise/ B_edges/ B_zstep/
C_area/ D_detect/ sentinels/

File name: <proc>_<target>_G<gap>_Z<zzzz>_F<field>_P<pose>_f<frame>.mc, for example
C_T5_G15_Z0800_F0_P017_f03.mc.

Manifest columns: file name; procedure; target ID; gap G (mm); station Z (mm); field position; pose index; frame index; random seed; commanded lateral offset (H, V mm); read-back robot pose (6 values); target pose in the camera frame from registration (6 values); timestamp; sensor temperature; air temperature; sensor-configuration ID.

Capture budget. The estimate below assumes FRAME_RATE_HZ = 10 frames/s and MOVE_AND_SETTLE_TIME_S = 3 s per move plus settle. Both are assumptions; the VSX3000 frame rate in the chosen trigger mode should replace them. Target swaps, warm-up, and the D post check are excluded. The plan tool recomputes the table from Section 2. Allow two to three working days in total.

Series	Poses	Frames	Robot time (h)
Registration	30	300	0.03
A main + tilt + sentinels	89	7,070	0.3
B-HV edges	520	15,600	0.87
B-Z ladder + ramp	453	4,530	0.5
C main + field	1,160	11,600	1.29
D main trials	2,160	2,160	1.86
D extended (low-point series)	2,880	2,880	2.48
Total	7,292	44,140	7.3

The extended low-point series takes about a third of the robot time. If time is short, 150 trials per configuration still measure the 5% point, to about $\pm 3.5\%$, and roughly halve that row. Check storage before starting: multiply the size of one .mc frame by about 44,000 frames.